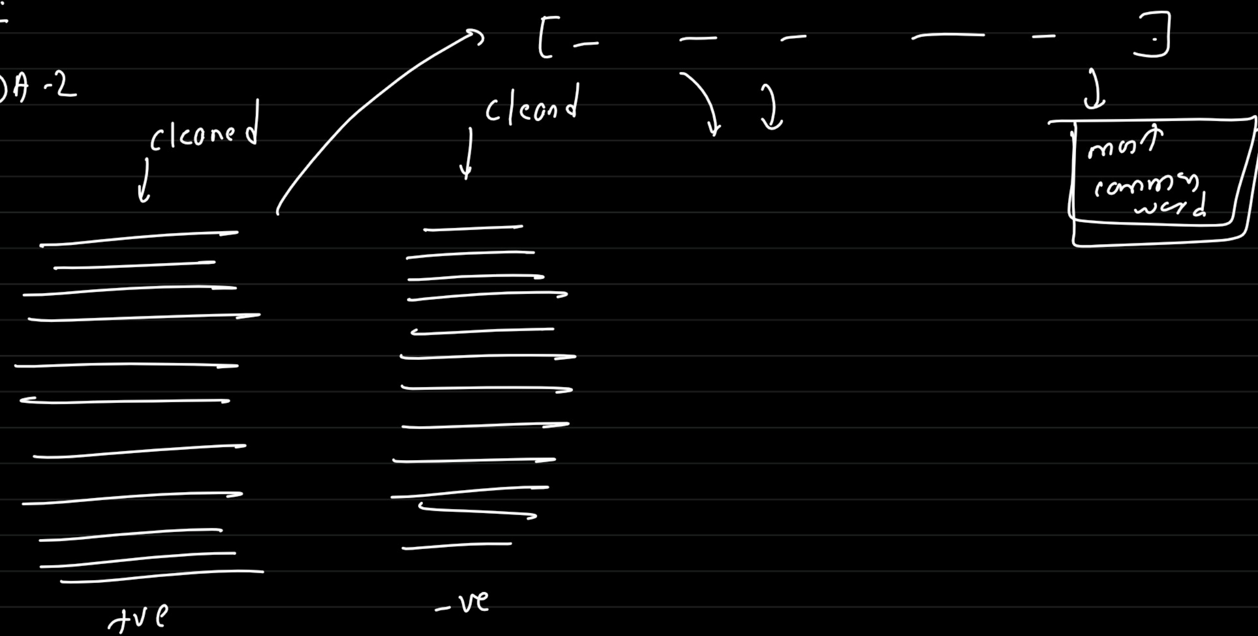
17-May-2026 (50th class)

Agenda:

EDA-2



funct([ ], top_n = 20)

Inferential stats

1. Hypothesis testing

binary case

Numerical → categorical [independent T-Test]

→ word_count → positive / negative

analysis of variance

comparing average between distinct group

> 2 cases → ANOVA

2

review long →
review short →

positive
negative

→ chi-square Test

$\frac{\text{class}}{\text{category}}$ → $\frac{\text{class}}{\text{category}}$

massive dataset ← sample

↓

a random meaningless difference in word count can yield a tiny p-value

→ outliers

random subsampling (smaller dataset) → realistic sample size

	-ve	+ve
catagery		
long	☐	☐
short	☐	☐

→ contingency

testing : whether knowing the category of one variable tells you anything about the category of other.

→ are they dependent or independent !

review

word-count

>100	<100
long	short

sentiment	
+ve	-ve

long → +ve → +1

long → -ve → +1

short → +ve → +1

short → -ve → +1

EDA → 100 → know more about the data

→ FE

X

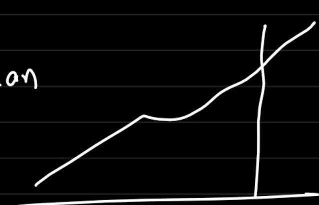
→ EDA → 2 → now

$\boxed{2,3}$

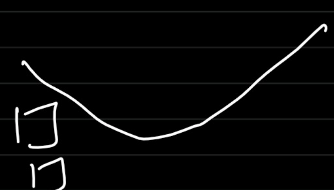
saturday →

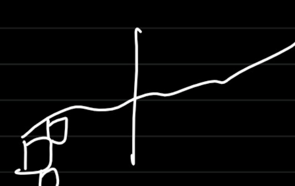
Time-series : —

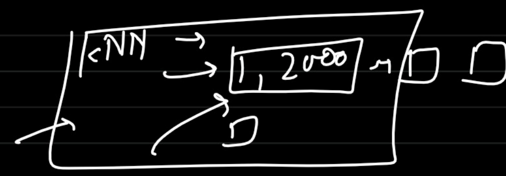
mean



→ 10 years → ?







→ ffill → production

	time	Gold
Jan		1,12,000
Feb		1,14,000
M		1,28,000
A		$\boxed{1,28,000}$
M		1,46,000

↑ ffill

ffill → forward fill → preceding valid observation to fill NAN

↓

data imputation